

PERTH PEARCE, Australia

WMO: 946120

Lat: 31.6669S

Lon: 116.0189E

Elev: 177

StdP: 14.60

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.9	41.1	27.5	21.6	74.4	30.5	24.8	71.9	25.5	61.2	23.0	60.6	2.4	0	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.5	101.1	67.1	97.2	66.7	94.2	66.3	73.3	87.8	70.9	86.6	69.3	85.3	12.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	108.8	78.1	66.4	97.8	74.8	64.6	91.6	73.9	37.0	88.5	34.9	86.7	33.5	84.8	97.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.7	21.3	19.5	DB	33.8	108.2	2.6	2.9	31.9	110.2	30.4	111.9	29.0	113.6	27.1	115.7	(4)
(5)			WB	32.9	76.7	2.8	4.8	30.9	80.2	29.3	83.0	27.8	85.7	25.8	89.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.7	77.0	77.5	74.0	67.7	61.8	57.4	55.3	55.9	57.8	62.5	68.7	73.1	(6)
(7)		DBStd	9.75	6.52	6.31	6.72	6.11	5.21	4.21	4.15	4.00	4.35	5.58	6.34	6.89	(7)
(8)		HDD50	12	0	0	0	0	0	2	7	2	1	0	0	0	(8)
(9)		HDD65	1363	1	1	6	39	126	229	300	284	220	119	30	8	(9)
(10)		CDD50	5728	837	769	745	532	365	224	172	184	235	387	560	718	(10)
(11)		CDD65	1604	372	350	286	121	26	2	0	1	5	41	140	260	(11)
(12)		CDH74	17758	4327	3983	3001	1052	207	4	0	8	66	467	1595	3048	(12)
(13)		CDH80	8653	2340	2106	1428	354	35	0	0	0	10	142	696	1542	(13)
(14)	Wind	WSAvg	9.3	11.2	11.1	10.3	8.3	7.7	8.0	8.1	8.6	9.3	10.5	11.0	(14)	
(15)	Precipitation	PrecAvg	26.0	0.2	0.5	0.6	1.3	3.5	5.4	5.3	4.1	2.7	1.5	0.9	0.3	(15)
(16)		PrecMax	38.2	1.5	3.8	3.4	4.5	7.7	11.4	9.6	7.1	5.4	3.3	3.7	1.5	(16)
(17)		PrecMin	18.1	0.0	0.0	0.0	0.0	0.2	0.8	2.1	1.2	0.3	0.1	0.0	0.0	(17)
(18)		PrecStd	4.9	0.4	0.8	0.7	1.0	1.7	2.1	1.9	1.3	1.3	0.8	0.9	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.9	104.8	102.2	93.2	84.0	73.8	71.3	74.7	80.6	90.3	98.5	104.2	(19)
(20)			MCWB	68.4	68.8	65.7	63.6	60.0	56.7	56.6	58.0	60.2	63.2	63.8	66.8	(20)
(21)		2%	DB	100.6	99.7	96.7	88.0	79.2	71.4	68.0	70.2	75.1	83.9	92.9	98.6	(21)
(22)			MCWB	67.6	68.1	67.1	63.0	59.2	57.2	56.5	57.6	59.0	61.9	63.6	65.9	(22)
(23)		5%	DB	96.6	95.6	91.8	84.1	75.4	68.4	65.9	67.6	71.0	79.0	87.8	93.4	(23)
(24)			MCWB	67.1	67.5	65.5	62.8	58.7	56.5	56.0	56.8	58.5	61.0	63.1	65.7	(24)
(25)		10%	DB	92.2	91.8	87.9	80.3	72.0	66.2	63.5	64.7	67.8	74.9	83.5	88.5	(25)
(26)			MCWB	66.6	66.9	65.3	61.9	58.6	55.8	55.3	55.7	57.3	60.2	62.0	64.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	75.5	73.4	69.3	66.6	63.4	62.2	62.7	63.7	66.9	69.2	83.3	(27)
(28)			MCD B	86.4	90.9	88.5	78.6	72.5	67.6	65.8	68.1	73.6	81.1	86.0	83.3	(28)
(29)		2%	WB	72.5	72.8	70.7	67.4	64.4	61.3	59.7	60.2	61.6	64.6	66.9	70.7	(29)
(30)			MCD B	88.2	88.6	84.0	77.8	70.7	66.0	63.8	65.6	70.1	77.4	83.1	86.7	(30)
(31)		5%	WB	70.3	70.6	69.0	65.8	62.8	59.6	58.1	58.8	60.0	62.9	65.3	68.1	(31)
(32)			MCD B	86.9	87.7	82.1	76.3	70.0	64.4	62.6	64.8	67.6	74.5	80.7	86.6	(32)
(33)		10%	WB	68.6	68.8	67.3	64.3	61.1	58.0	56.6	57.2	58.5	61.3	63.8	66.3	(33)
(34)			MCD B	86.0	86.2	81.3	75.3	68.3	63.0	61.4	62.9	65.8	71.7	78.4	84.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	27.3	26.5	24.8	23.0	20.5	17.7	16.9	18.2	19.6	23.0	25.6	26.7	(35)
(36)			MCDBR	32.8	31.2	29.4	27.3	24.9	20.3	18.8	22.0	25.4	29.2	32.6	33.3	(36)
(37)			MCWBR	10.4	9.8	9.6	10.5	11.6	11.6	11.1	12.9	13.6	12.9	11.7	10.6	(37)
(38)		5% WB	MCDBR	27.2	26.2	23.8	21.4	18.7	16.3	15.0	17.4	21.2	24.8	28.3	28.9	(38)
(39)			MCWBR	10.4	9.8	9.6	10.3	11.1	11.4	10.6	11.3	12.6	12.3	11.6	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.340	0.342	0.325	0.316	0.306	0.297	0.296	0.304	0.325	0.327	0.328	0.329	(40)	
(41)		taud	2.558	2.569	2.605	2.616	2.616	2.625	2.624	2.594	2.510	2.526	2.548	2.564	(41)	
(42)		Ebn at Noon	317	310	305	292	279	274	280	292	300	312	319	321	(42)	
(43)		Edh at Noon	34	33	30	26	23	22	23	26	32	34	34	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2604	2350	1948	1433	1077	884	935	1207	1604	2081	2470	2668	(44)	
(45)		RadStd	122	81	97	82	72	62	56	57	89	94	102	113	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +0.92	(e) N/A	(f) N/A	(g) +2.11	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +349	(m) N/A	(46)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page